Daily Number Theory Drill #6

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: The Euclidean Step · Polynomial Division · GCD Logic.

Section A

1. Find the largest prime factor of 2025.

Ans: 5

2. What is the last digit of 3^{2026}?

Ans: 9

3. Find the greatest common divisor of 84 and 120.

Ans: 12

4. What is the remainder when 10^{10} is divided by 9?

Ans: 1

5. Find the missing digit d such that the five-digit number $47d53$ is divisible by 9.

Ans: 8

6. How many positive integer factors does 100 have?

Ans: 9

7. Which is larger: 2^{300} or 3^{200}?

Ans: 3^{200}

8. Evaluate $5^{2024} \pmod 4$.

Ans: 1

9. Find the smallest positive integer n such that $12n$ is a perfect square.

Ans: 3

10. What is the sum of the prime factors of 210?

Ans: 17

Section B

1. Find the number of trailing zeroes of 20!.

Ans: 4

2. Solve for x in $3x \equiv 4 \pmod 7$, with $0 < x < 7$.

Ans: 6

3. If p and q are prime numbers such that $p^2 - q^2 = 24$, find the value of $p + q$.

Ans: 12

4. Find the last two digits of 5^{2025}.

Ans: 25

5. A number N leaves a remainder of 2 when divided by 3, and 3 when divided by 5. What is the smallest positive integer value of N?

Ans: 8

6. How many perfect squares divide 3600?

Ans: 12

7. Find the highest power of 3 dividing $27^5 \times 9^4$.

Ans: 23

8. What is the smallest positive integer with exactly 6 positive divisors?

Ans: 12

9. If x and y are positive integers such that $xy = 144$ and $\gcd(x, y) = 3$, find the maximum possible value of $x + y$.

Ans: 51

10. Calculate the remainder when $1! + 2! + 3! + \cdots + 10!$ is divided by 5.

Ans: 3

Section C

1. Find the number of pairs of positive integers (x, y) such that $xy + x + y = 23$.

Ans: 6

2. Find the sum of all prime numbers p such that $p^2 + 2$ is also prime.

Ans: 3

3. How many 3-digit palindromes are divisible by 11?

Ans: 8

4. The product of three consecutive positive integers is equal to 336. Find the sum of these integers.

Ans: 21

5. Let N be a 4-digit number. When its digits are reversed, the new number is $N + 1089$. If the sum of the digits of N is 12, what is the maximum possible value of N?

Ans: 5016

6. Find the sum of all positive integers x such that $\frac{x+11}{x-3}$ is an integer.

Ans: 39

7. Find the maximum possible value of $\gcd(7n + 15, 3n + 2)$, where n is a positive integer.

Ans: 31

8. Find the sum of all integers n such that $\frac{n^2+3n+5}{n-1}$ is an integer.

Ans: 6

Section D

1. Find all three-digit numbers abc such that $abc = a! + b! + c!$.

Ans: 145

2. Let $d_n = \gcd(n^2 + 3, n + 1)$. Find the sum of all possible values of d_n as n ranges over all positive integers.

Ans: 7

3. Find the last three digits of 2025^{2025}.

Ans: 625

4. For how many positive integers $n \leq 100$ is n^n a perfect square?

Ans: 55

5. Find the maximum value of $x+y+z$ given that x, y, z are positive integers satisfying $x \leq y \leq z$ and $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=1$.

Ans: 11